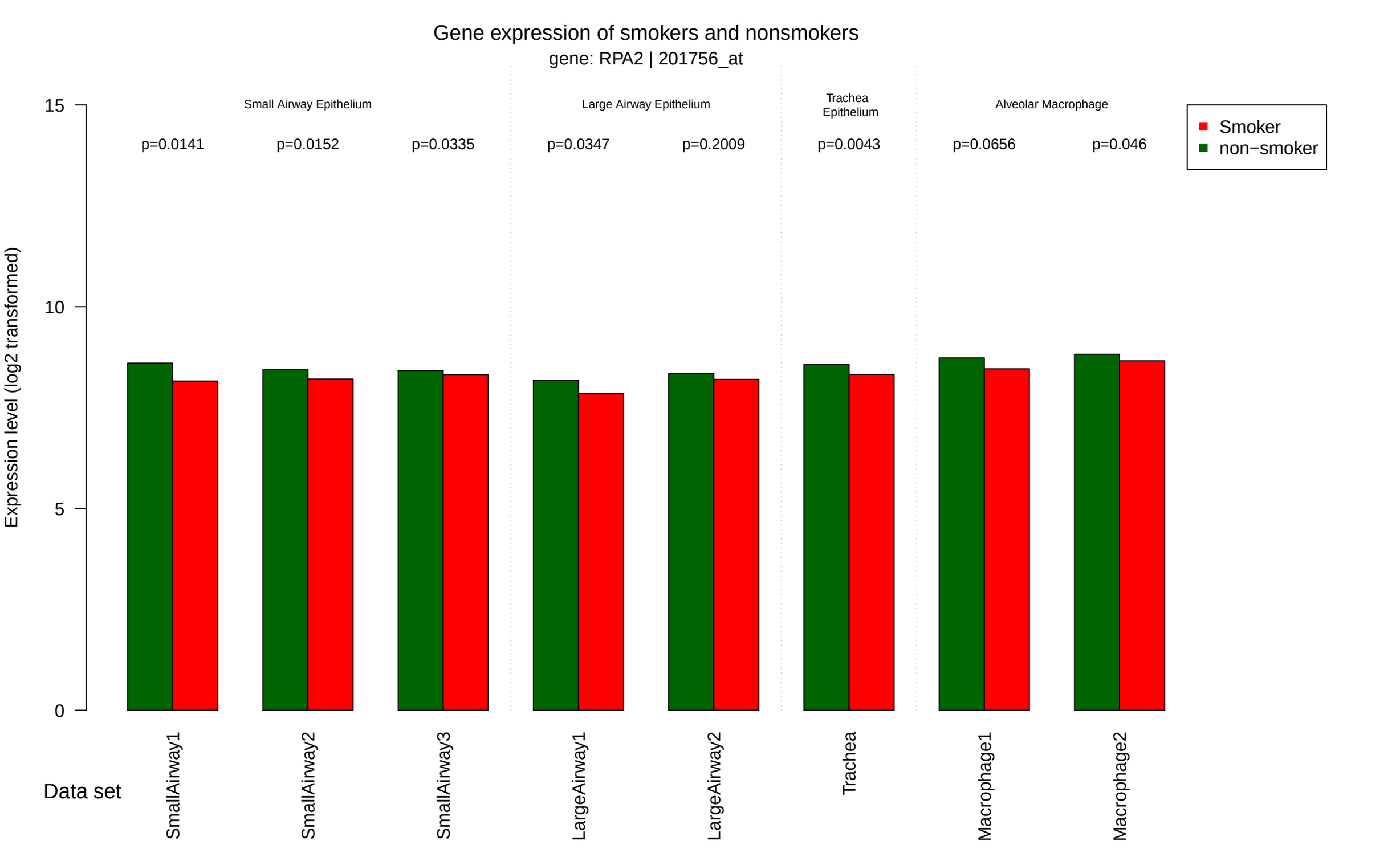
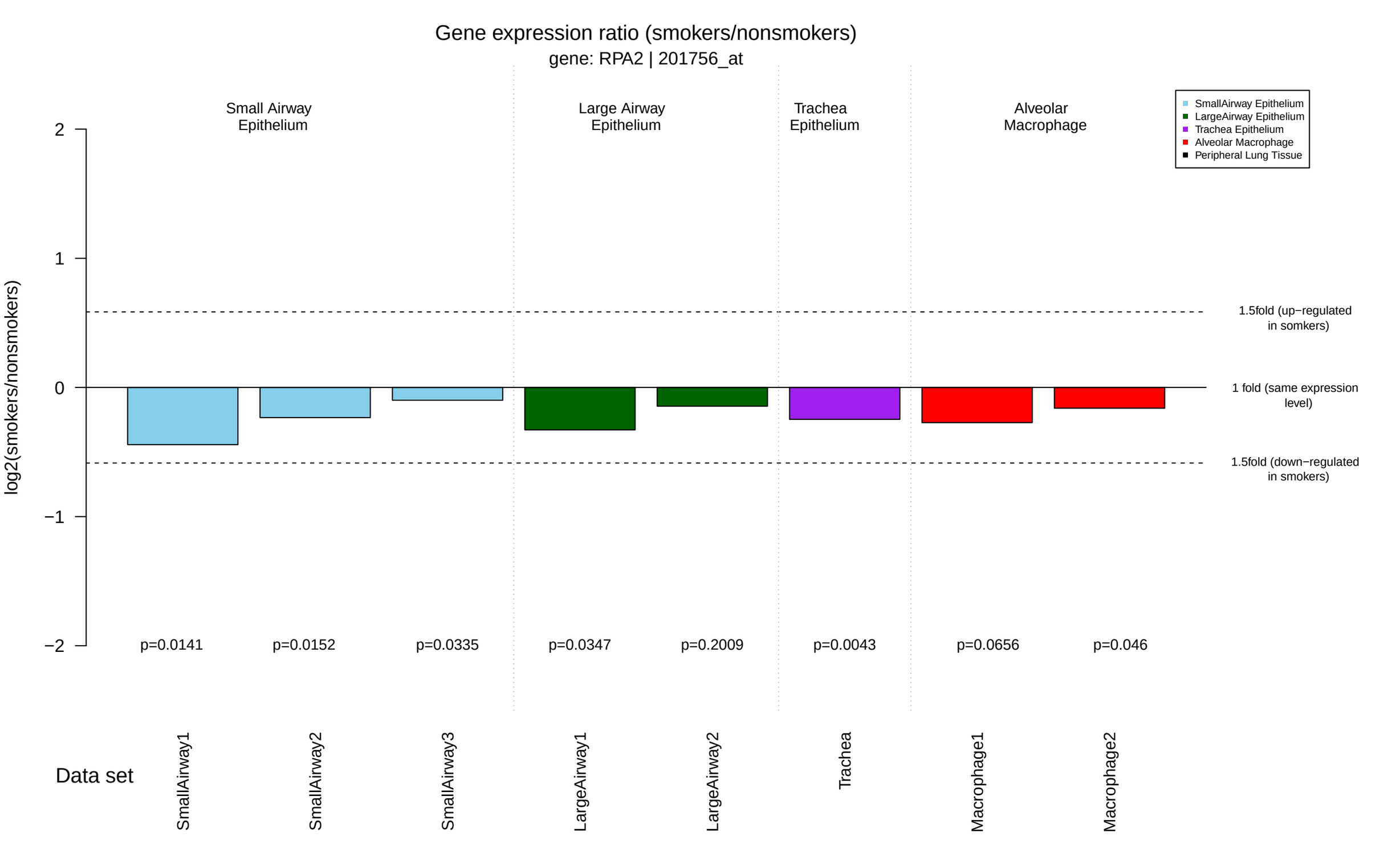
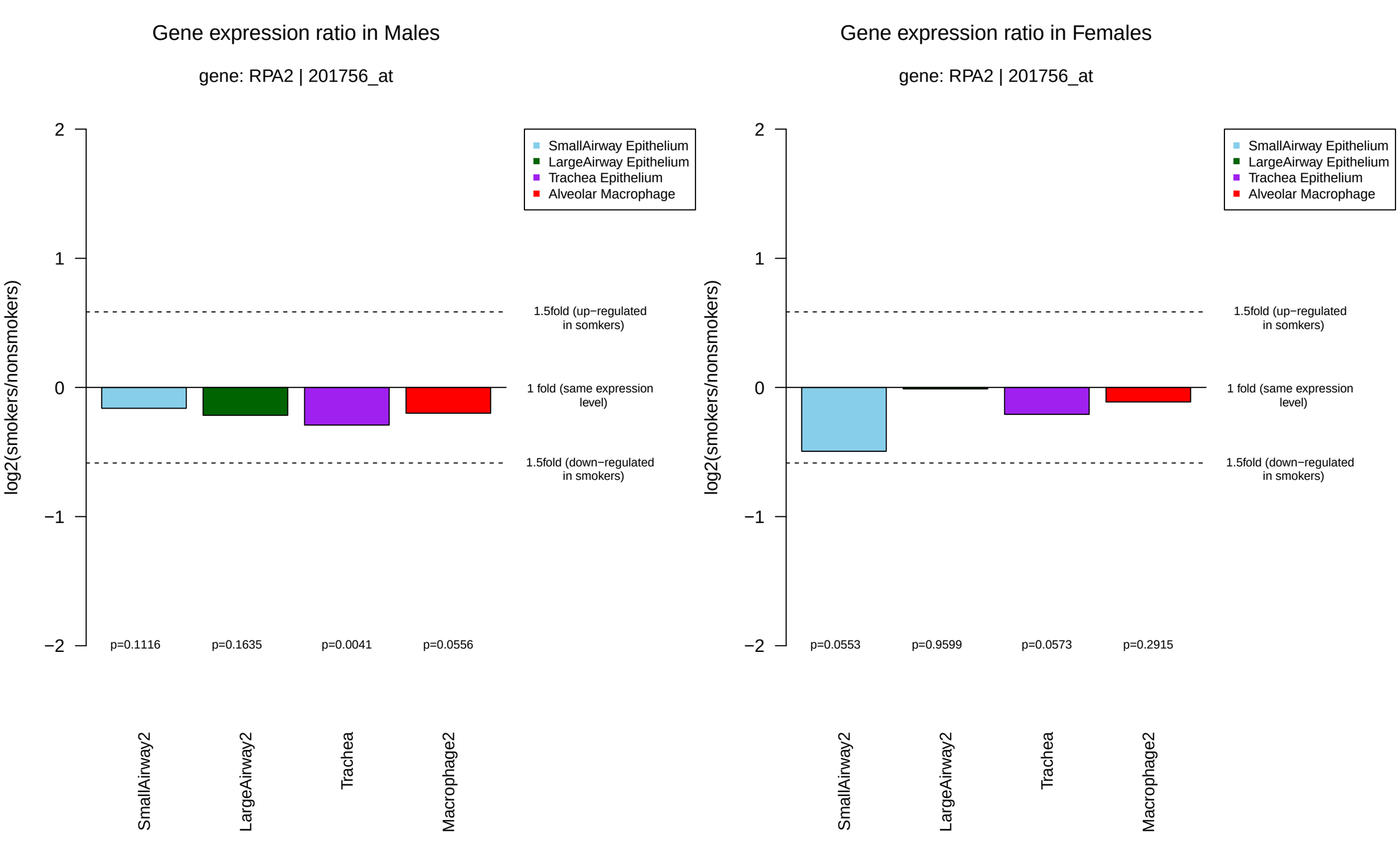
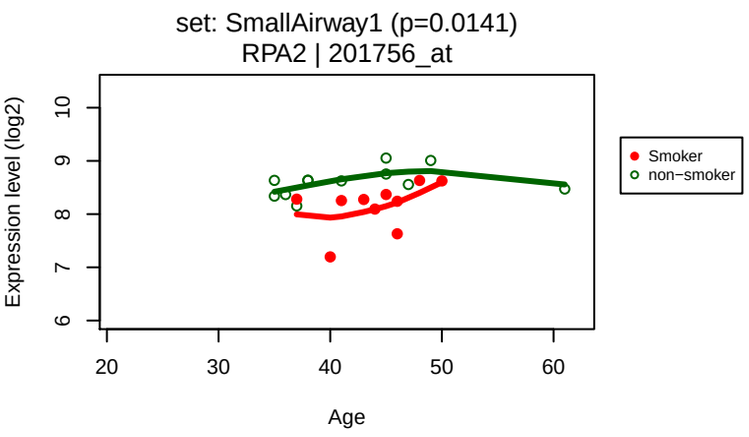
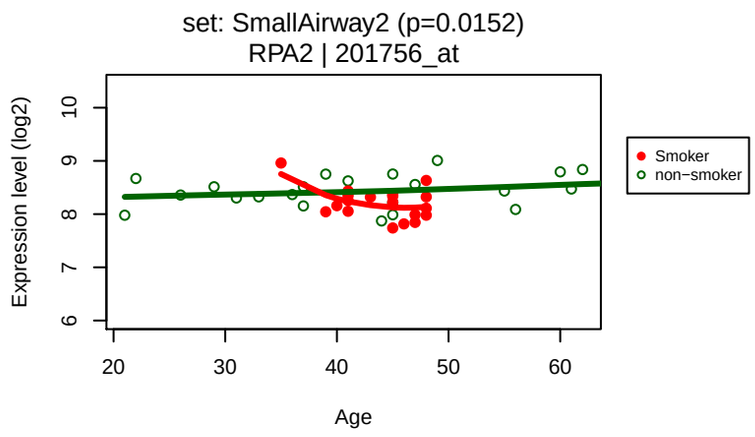
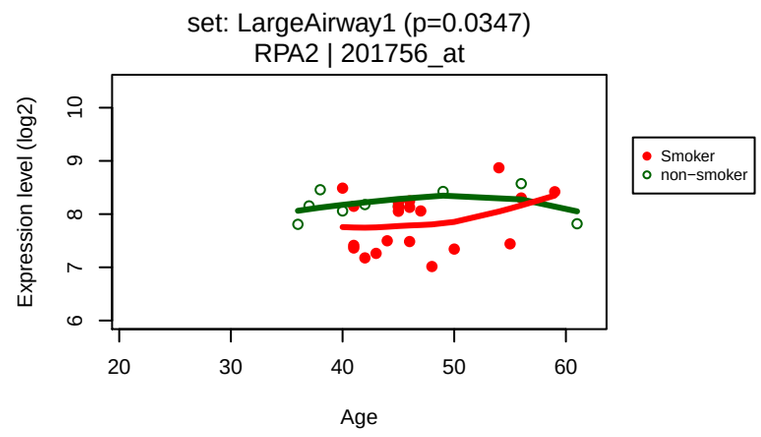

Gene expression of smokers and nonsmokers
gene: RPA2 | 201756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0141
p=0.0152
p=0.0335
p=0.0347
p=0.2009
p=0.0043
p=0.0656
p=0.046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

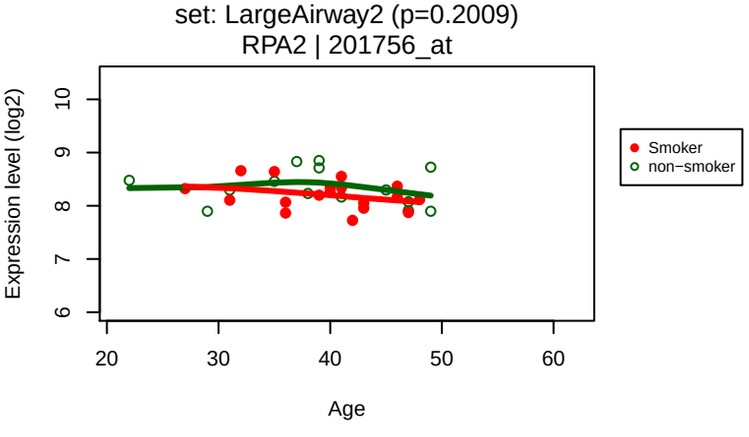

Gene expression ratio (smokers/nonsmokers)
gene: RPA2 | 201756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0141
p=0.0152
p=0.0335
p=0.0347
p=0.2009
p=0.0043
p=0.0656
p=0.046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

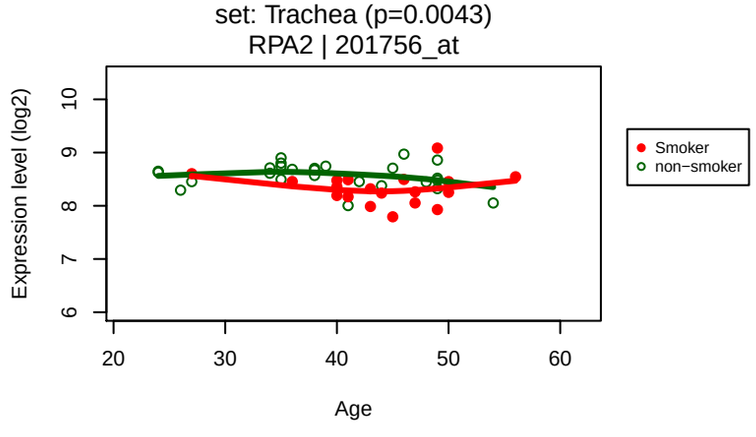

Gene expression ratio in Males
gene: RPA2 | 201756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1116
p=0.1635
p=0.0041
p=0.0556
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPA2 | 201756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0553
p=0.9599
p=0.0573
p=0.2915
SmallAirway2
LargeAirway2
Trachea
Macrophage2

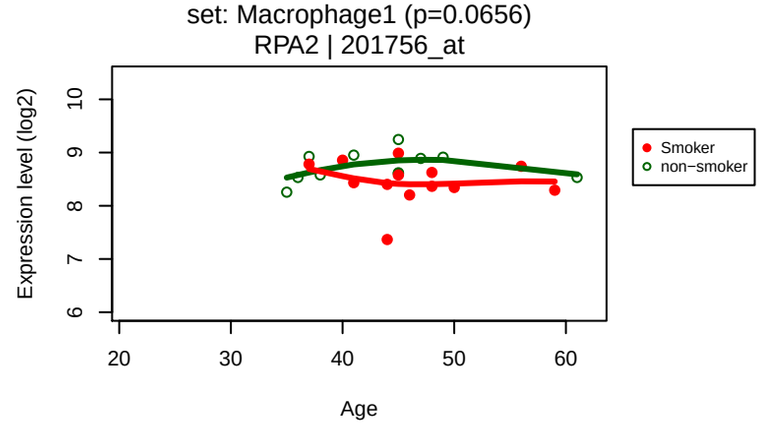

set: SmallAirway1 (p=0.0141)
RPA2 | 201756_at
Expression level (log2)
Age
Smoker
non−smoker

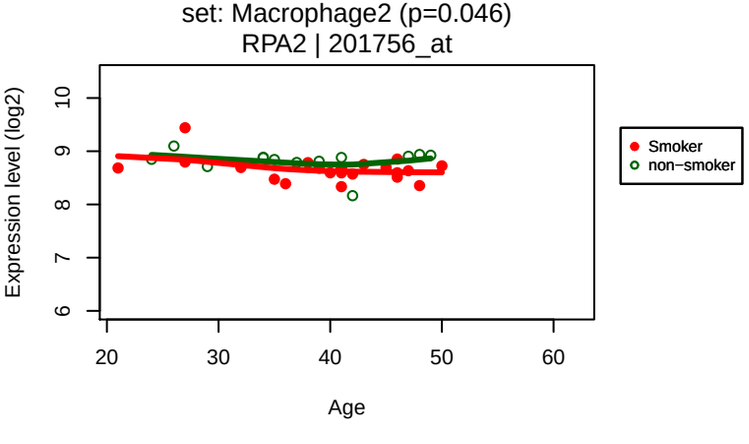

set: SmallAirway2 (p=0.0152)
RPA2 | 201756_at
Expression level (log2)
Age
Smoker
non−smoker

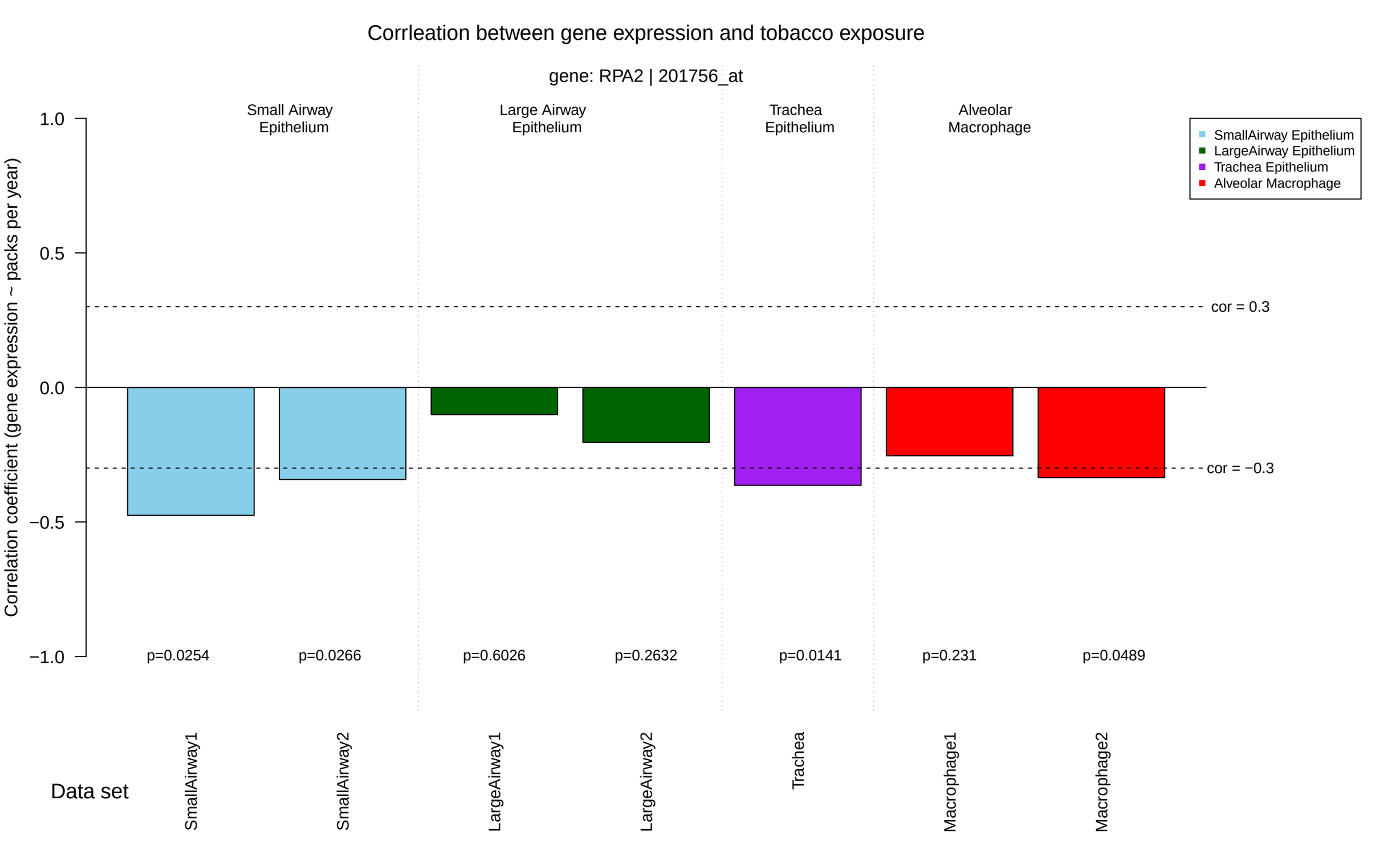

set: LargeAirway1 (p=0.0347)
RPA2 | 201756_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2009)
RPA2 | 201756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0043)
RPA2 | 201756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0656)
RPA2 | 201756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.046)
RPA2 | 201756_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPA2 | 201756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0254
p=0.0266
p=0.6026
p=0.2632
p=0.0141
p=0.231
p=0.0489
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2